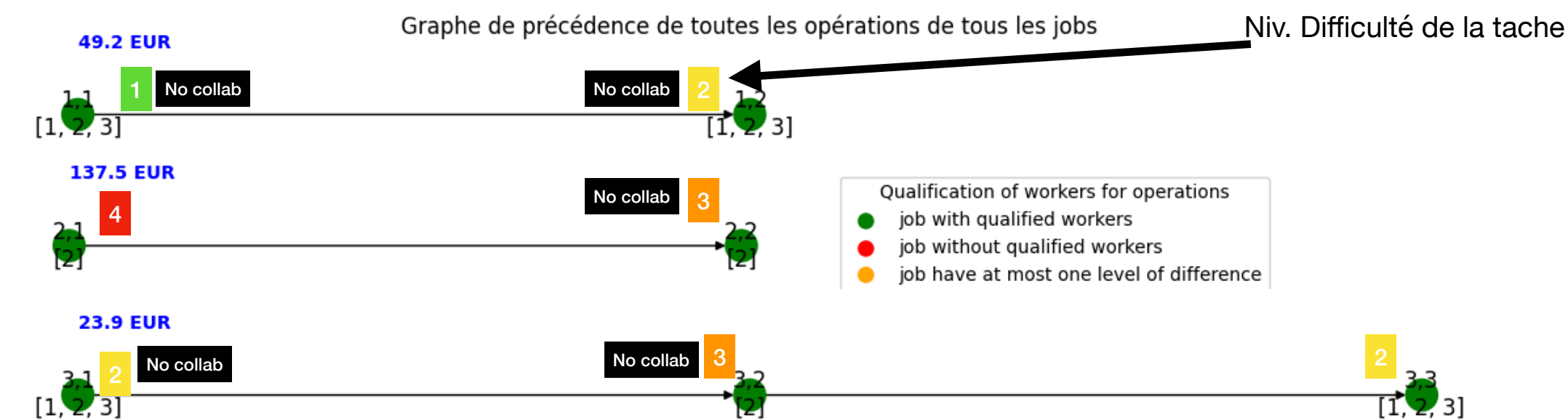
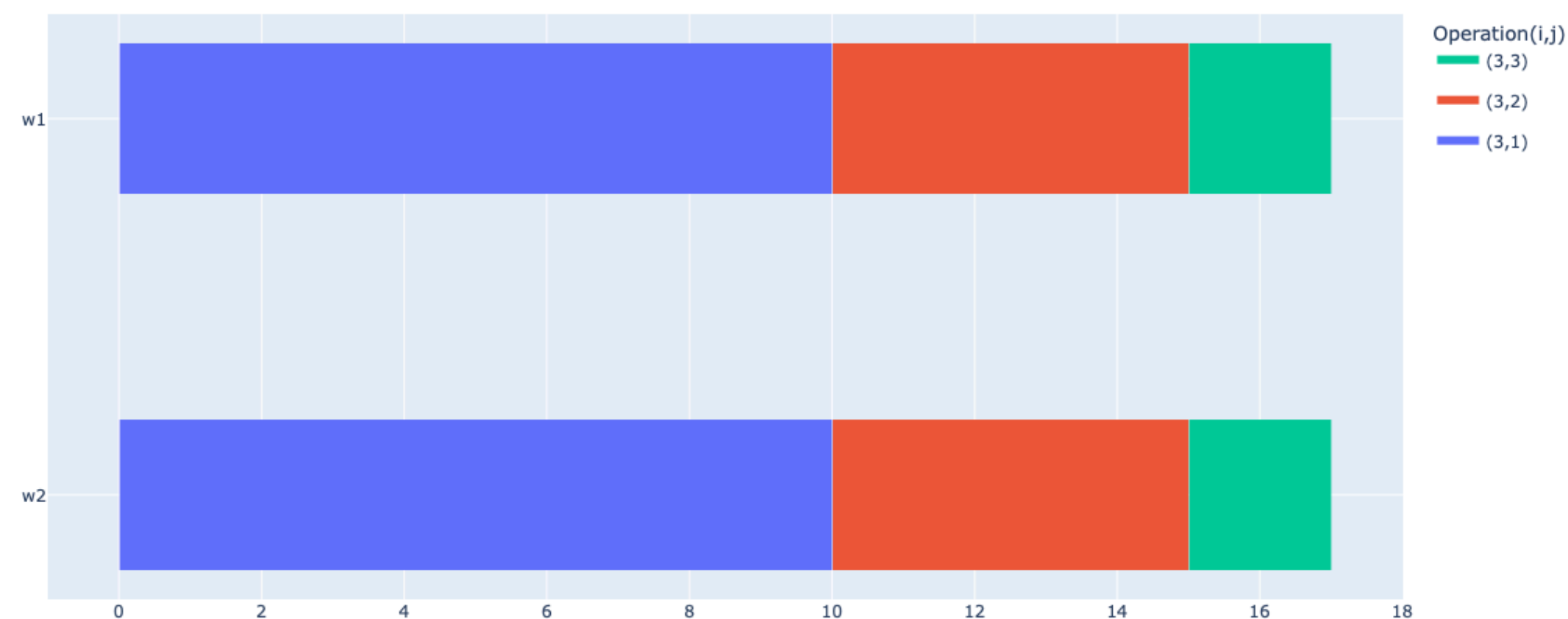


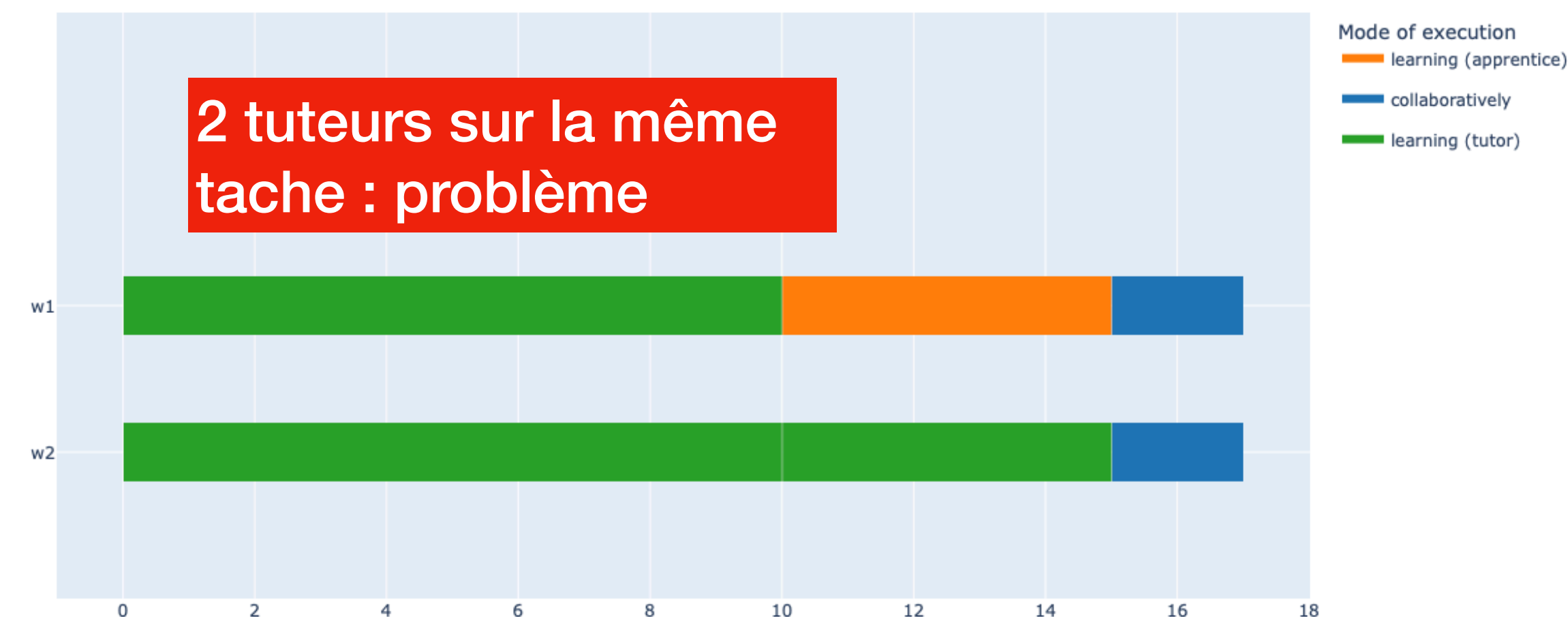


TIME_LIMIT + 19.5 et No solo task actif

Gantt Chart (makespan= 17.0)



Gantt Chart (makespan= 17.0)



Linéarisation du min

$$level_min_{ij} \leq x_{ijk} \cdot L_{km} + M \cdot (1 - x_{ijk}) \quad \forall i \in J, j \in J_i, k \in W \quad (6)$$

$$level_min_{ij} \geq x_{ijk} \cdot L_{km} - M \cdot (1 - \delta'_{ijk}) \quad \forall i \in J, j \in J_i, k \in W \quad (7)$$

$$\sum_{k \in W} \delta'_{ijk} \leq 1 \quad \forall i \in J, j \in J_i \quad (8)$$

$$\delta'_{ijk} \leq x_{ijk} \quad \forall i \in J, j \in J_i, k \in W \quad (9)$$

Ajout de la contrainte suivante: $\sum_k \delta'_{ijk} \geq task_done_{ij} \quad \forall i \in J, j \in J_i$

Pour forcer a choisir l'indice d'un worker k

Correction du pb trouvé

Après ajout de la contrainte manquante :

```
===== Checking which jobs are completed: =====
completed jobs: []
```

Pas de job réalisé

```
Worker levels: shape=(3, 1)
[[2.45068583]
 [4.         ]
 [2.11387086]]
```

Logique car

Job 1 : tout le monde à le niveau mais possible en collab

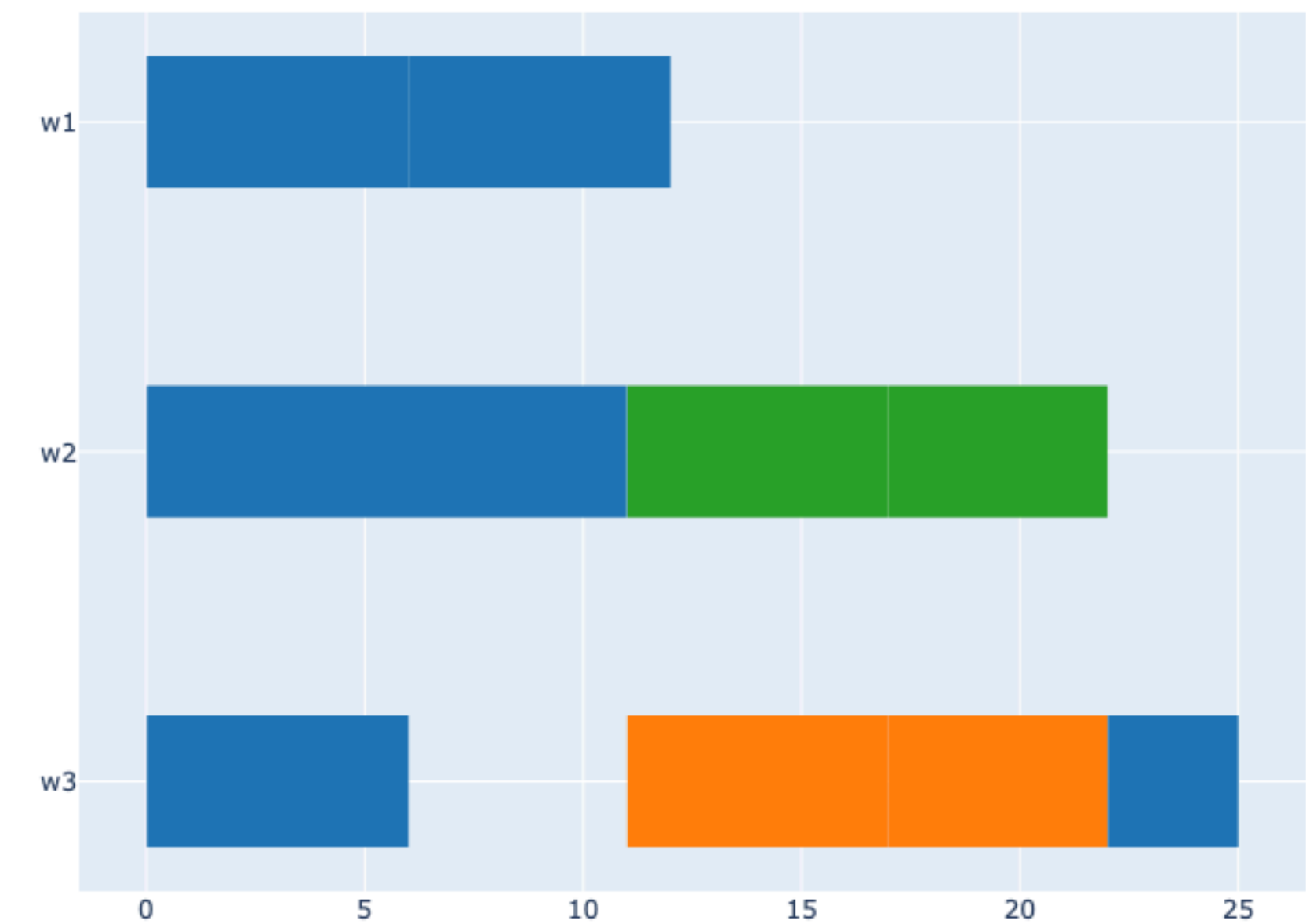
Job 2 : la premiere opération w2 peut l'enseigner car au moins diff de 1 pour l'apprentis

Job 3 : tout le monde à le niveau pour la premiere opération mais pas de collab possible

```
#===== Dico de pondération des objectifs (Aggrégation) =====
weights = {
    "profit" : 100,
    "skills" : 100,
    "cognitive_load" : 1
}
```

En enlevant la contrainte de no task solo

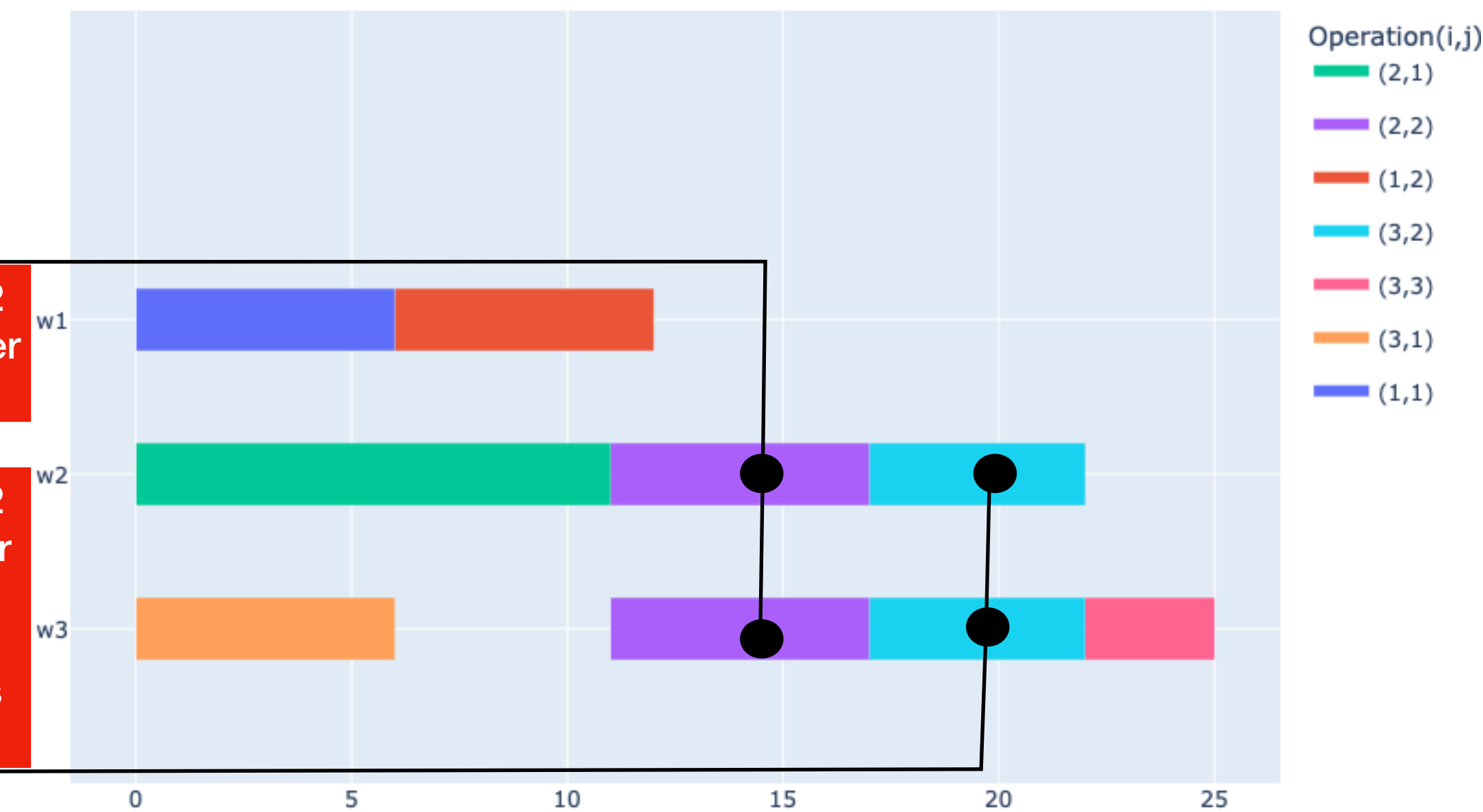
Gantt Chart (makespan= 25.0)



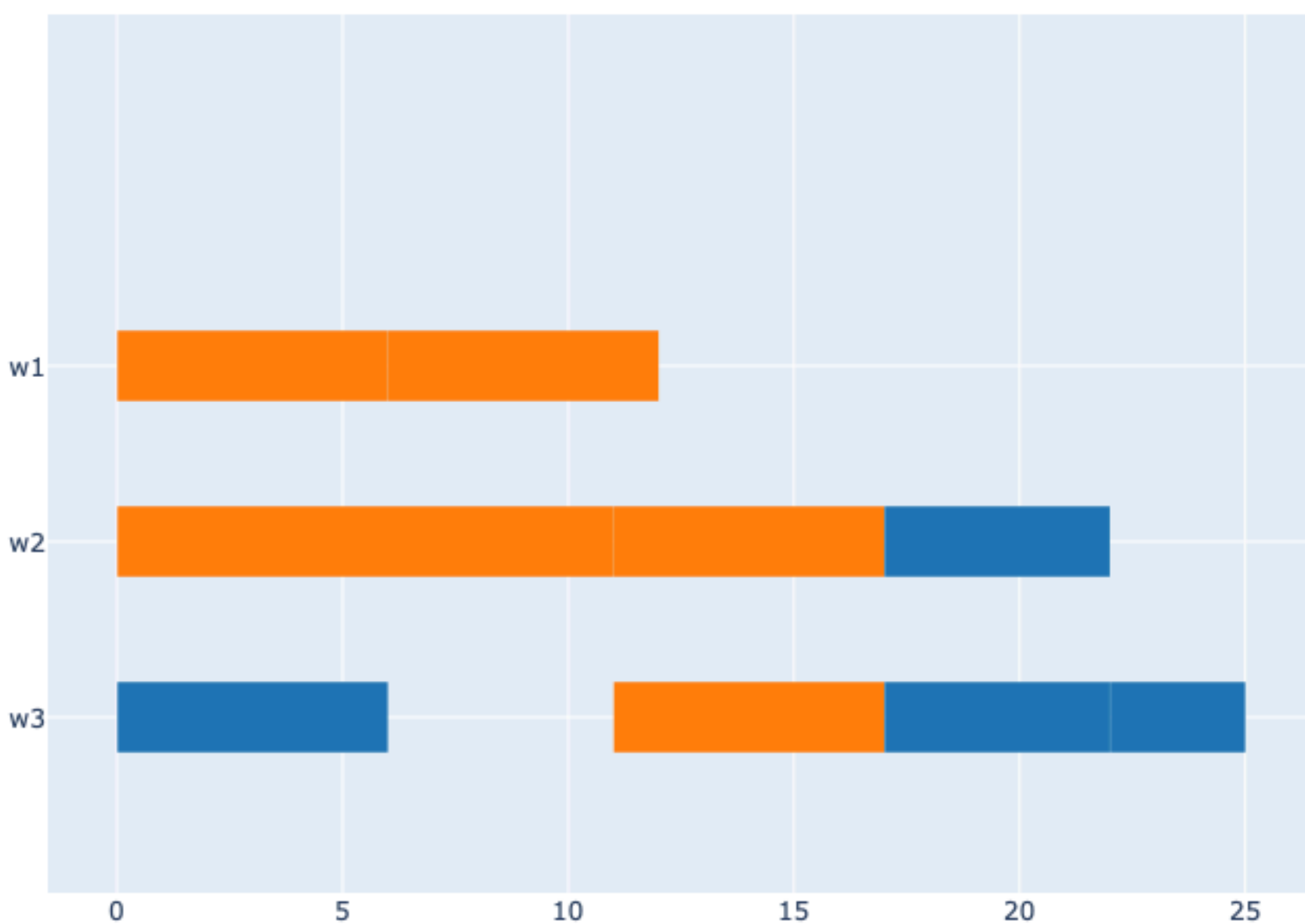
w2 peut faire la tache 22
Seul mais pour augmenter
skills l'apprend a w3

w2 peut faire la tache 32
Seul mais comme worker
de base est w3, w3
demande de l'aide a w2
pour 32 car w3 peut pas
la faire seul

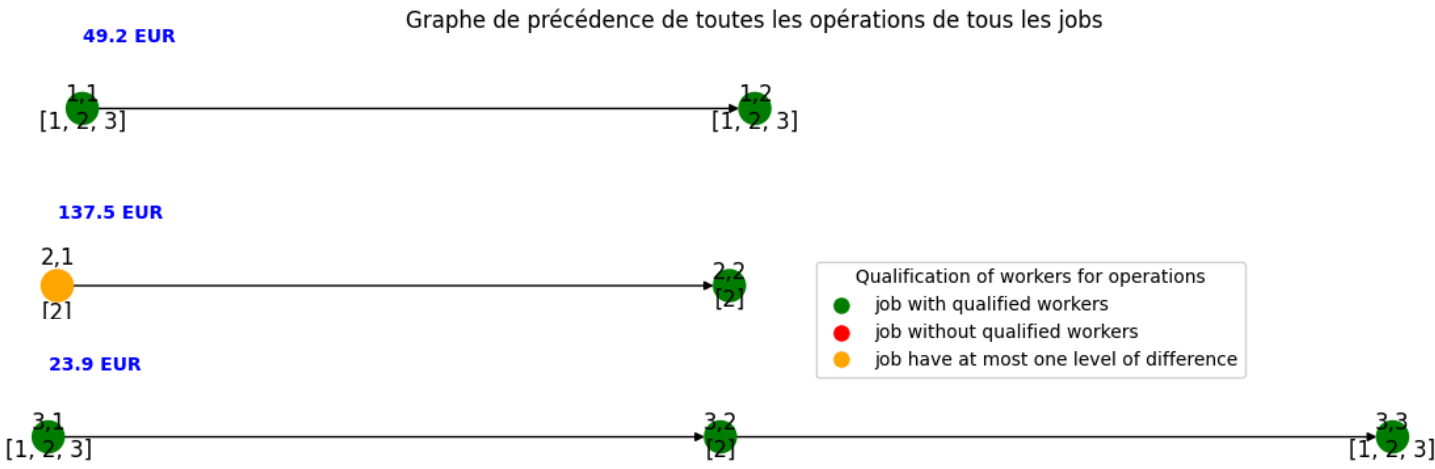
Gantt Chart (makespan= 25.0)



Gantt Chart (makespan= 25.0)



seed = 135



Time limit = 10

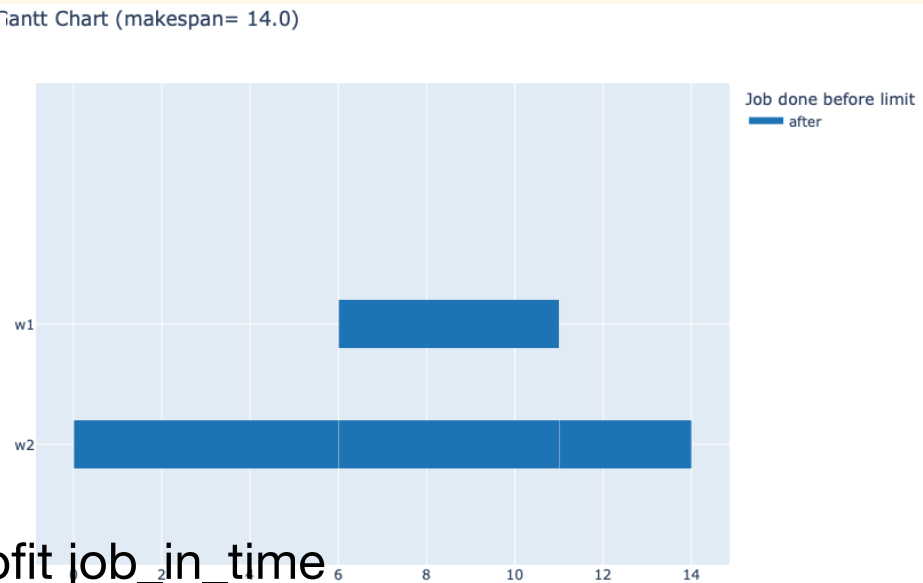
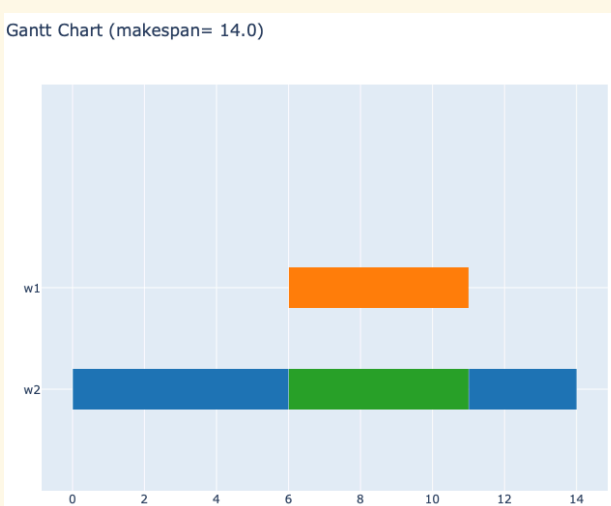
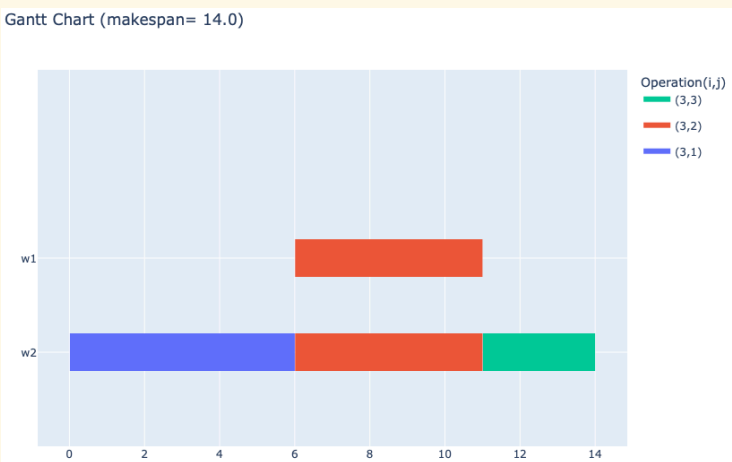
`obj_benefit = benefit_obj_in_time - penalty_levels_obj - penalty_makespan_job_OBJ`

=====
completed jobs: []

`obj_benefit = benefit_obj_in_time - penalty_levels_obj`

`obj_benefit = benefit_obj_in_time`

Pareil car solo sans level pas autorisé



Objective value: 29.02651911367218

N'a aucun interet car Max profit job_in_time

Rq : Le job1 peut etre fait apres mais ne prefere pas l'ordonnancer

Autorisation solo sans level

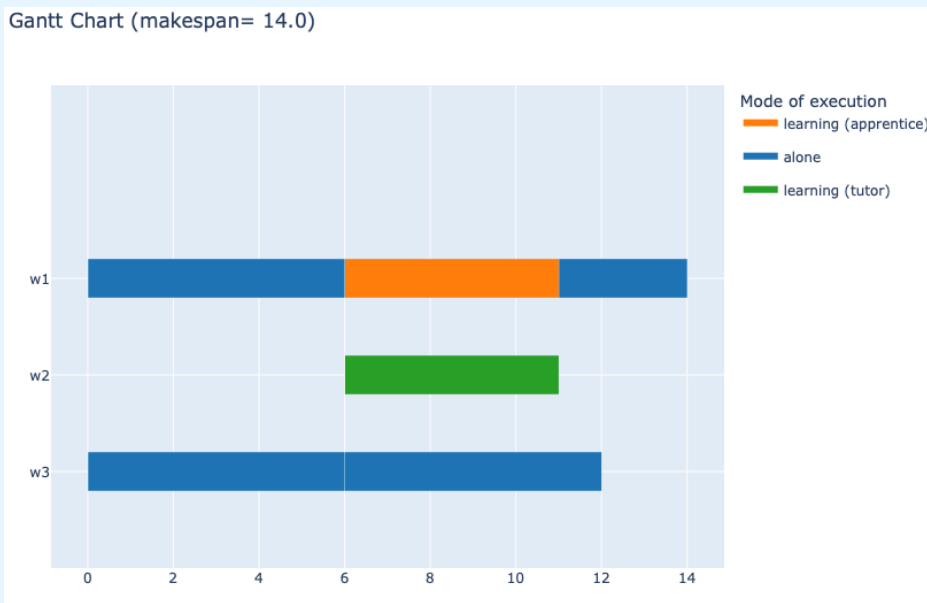
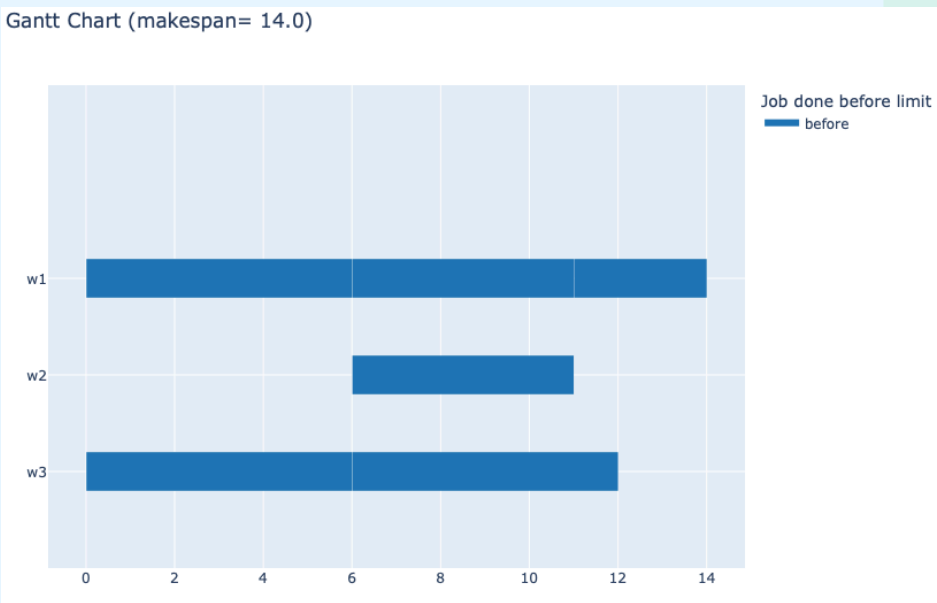
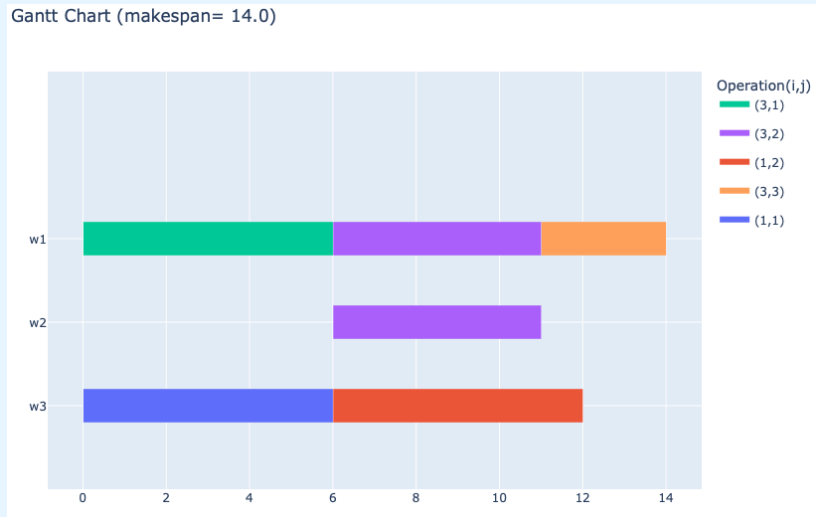
Time limit = 19.5

`obj_benefit = benefit_obj_in_time - penalty_levels_obj - penalty_makespan_job_OBJ`

`obj_benefit = benefit_obj_in_time - penalty_levels_obj`

`obj_benefit = benefit_obj_in_time`

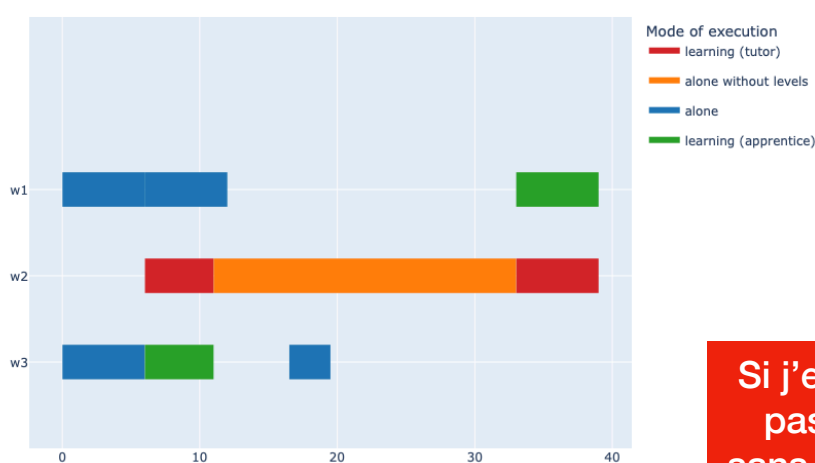
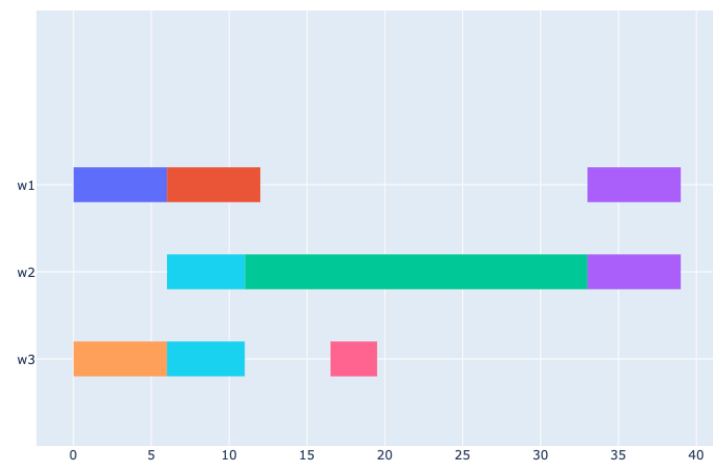
Donne tout pareil



Time limit = 19.5

Objective value: 7342.7297172833305

`obj_benefit = benefit_obj_in_time`

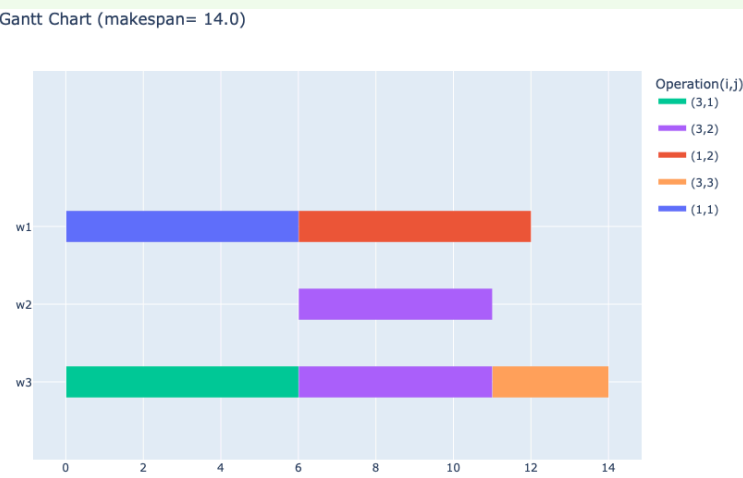


7410.486784418428

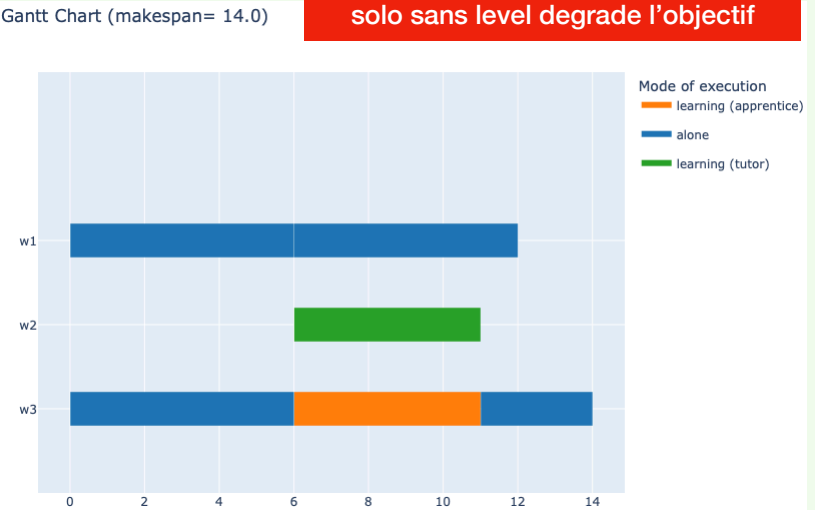
`obj_benefit = benefit_obj_in_time - penalty_levels_obj`

Signification: Laisser faire la tache en solo sans level degrade l'objectif

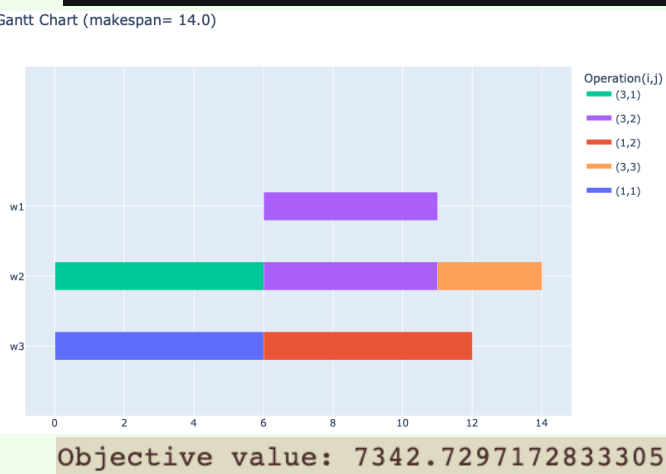
Si j'enlève ce terme il se peut que l'on est pas besoin qu'un worker fasse en solo sans level mais il pourra quand même faire (pb si on veut vraiment en dernier recours)



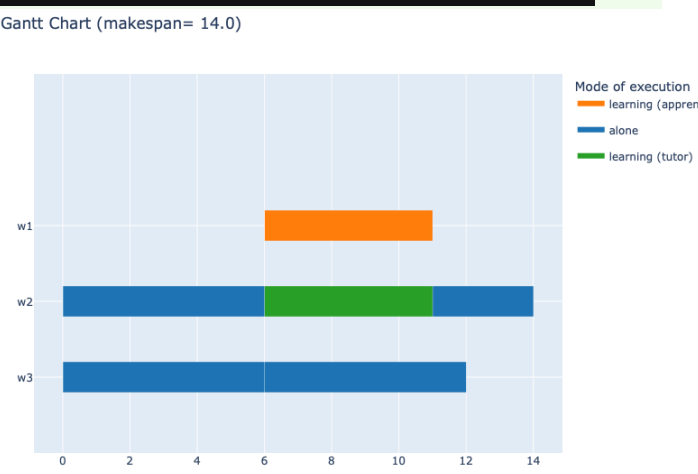
Objective value: 7342.460265304756



`obj_benefit = benefit_obj_in_time - penalty_levels_obj - penalty_makespan_job_OBJ`



Objective value: 7342.7297172833305

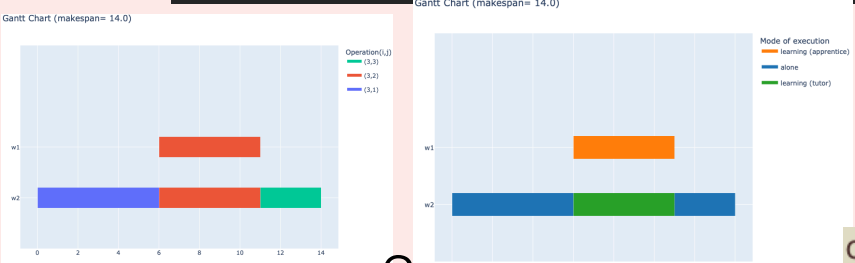


Time limit = 10

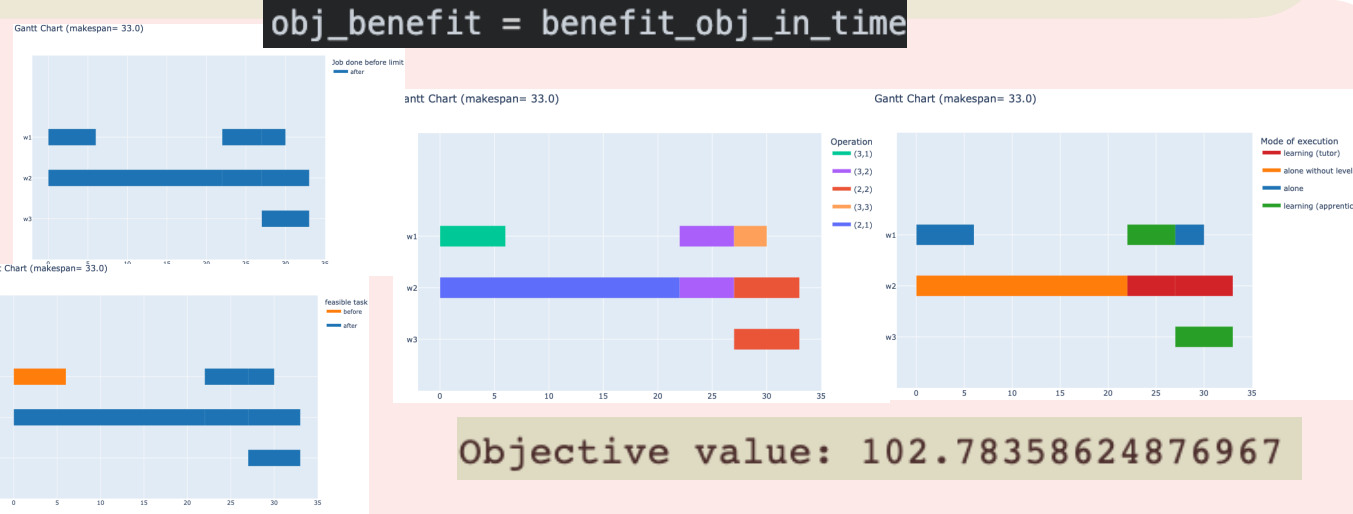
`obj_benefit = benefit_obj_in_time - penalty_levels_obj - penalty_makespan_job_OBJ`

=====
completed jobs: []

`obj_benefit = benefit_obj_in_time - penalty_levels_obj`



Objective value: 29.02651911367218



Objective value: 102.78358624876967